

```
SET SQL_SAFE_UPDATES = 0;
```

```
-- Module: Database Design and  
Introduction to SQL
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-- Session: Database Creation in MySQL  
Workbench
```

```
-- DDL Statements
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```
-- 1. Create a table shipping_mode_dimen  
having columns with their respective data  
types as the following:
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```
--      (i) Ship_Mode VARCHAR(25)
```

```
--      (ii) Vehicle_Company VARCHAR(25)
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```
--      (iii) Toll_Required BOOLEAN
```

```
-- 2. Make 'Ship_Mode' as the primary key  
in the above table.
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-- DML Statements
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-- 1. Insert two rows in the table  
created above having the row-wise values:
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```
--      (i) 'DELIVERY TRUCK', 'Ashok  
Leyland', false
```

```
--      (ii) 'REGULAR AIR', 'Air India',  
false
```

```
-- 2. The above entry has an error as  
land vehicles do require tolls to be  
paid. Update the 'Toll_Required'  
attribute  
-- to 'Yes'.
```

```
-- 3. Delete the entry for Air India.
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-- Adding and Deleting Columns
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-- 1. Add another column named  
'Vehicle_Number' and its data type to the  
created table.
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-- 2. Update its value to 'MH-05-R1234'.
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```
-- 3. Delete the created column.
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-- Changing Column Names and Data Types
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-- 1. Change the column name  
'Toll_Required' to 'Toll_Amount'. Also,  
change its data type to integer.
```

-- 2. The company decides that this additional table won't be useful for data analysis. Remove it from the database.

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-- Session: Querying in SQL
-- Basic SQL Queries

-- 1. Print the entire data of all the customers.

-- 2. List the names of all the customers.

-- 3. Print the name of all customers along with their city and state.

-- 4. Print the total number of customers.
select count(*) as Total_Customers
from cust_dimen;

-- 5. How many customers are from West Bengal?

-- 6. Print the names of all customers who belong to West Bengal.

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-- Operators

-- 1. Print the names of all customers who are either corporate or belong to Mumbai.

-- 2. Print the names of all corporate customers from Mumbai.

-- 3. List the details of all the customers from southern India: namely Tamil Nadu, Karnataka, Telangana and Kerala.

-- 4. Print the details of all non-small-business customers.

-- 5. List the order ids of all those orders which caused losses.

-- 6. List the orders with '_5' in their order ids and shipping costs between 10 and 15.

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-- Aggregate Functions

-- 1. Find the total number of sales made.

-- 2. What are the total numbers of customers from each city?

-- 3. Find the number of orders which have been sold at a loss.

-- 4. Find the total number of customers from Bihar in each segment.

-- 5. Find the customers who incurred a shipping cost of more than 50.

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-- Ordering

-- 1. List the customer names in alphabetical order.

-- 2. Print the three most ordered products.

-- 3. Print the three least ordered products.

-- 4. Find the sales made by the five most profitable products.

-- 5. Arrange the order ids in the order of their recency.

-- 6. Arrange all consumers from Coimbatore in alphabetical order.

--

-- String and date-time functions

-- 1. Print the customer names in proper case.

-- 2. Print the product names in the following format: Category_Subcategory.

-- 3. In which month were the most orders shipped?

-- 4. Which month and year combination saw the most number of critical orders?

-- 5. Find the most commonly used mode of shipment in 2011.

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-- Regular Expressions

-- 1. Find the names of all customers having the substring 'car'.

-- 2. Print customer names starting with A, B, C or D and ending with 'er'.

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-- Nested Queries

-- 1. Print the order number of the most valuable order by sales.

-- 2. Return the product categories and subcategories of all the products which don't have details about the product
-- base margin.

-- 3. Print the name of the most frequent customer.

-- 4. Print the three most common products.

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-- CTEs

-- 1. Find the 5 products which resulted
in the least losses. Which product had
the highest product base
-- margin among these?

-- 2. Find all low-priority orders made
in the month of April. Out of them, how
many were made in the first half of
-- the month?

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-- Views

-- 1. Create a view to display the sales
amounts, the number of orders, profits
made and the shipping costs of all
-- orders. Query it to return all orders
which have a profit of greater than 1000.

-- 2. Which year generated the highest
profit?

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-- Session: Joins and Set Operations

-- Inner Join

-- 1. Print the product categories and subcategories along with the profits made for each order.

-- 2. Find the shipment date, mode and profit made for every single order.

-- 3. Print the shipment mode, profit made and product category for each product.

-- 4. Which customer ordered the most number of products?

-- 5. Selling office supplies was more profitable in Delhi as compared to Patna. True or false?

-- 6. Print the name of the customer with the maximum number of orders.

-- 7. Print the three most common products.

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-- Outer Join

-- 1. Return the order ids which are
present in the market facts table.

-- Execute the below queries before
solving the next question.
create table manu (
    Manu_Id int primary key,
    Manu_Name varchar(30),
    Manu_City varchar(30)
);

insert into manu values
(1, 'Navneet', 'Ahemdabad'),
(2, 'Wipro', 'Hyderabad'),
(3, 'Furlanco', 'Mumbai');

alter table Prod_Dimen
add column Manu_Id int;

update Prod_Dimen
set Manu_Id = 2
where Product_Category = 'technology';

-- 2. Display the products sold by all
the manufacturers using both inner and
outer joins.

```

-- 3. Display the number of products sold by each manufacturer.

-- 4. Create a view to display the customer names, segments, sales, product categories and

-- subcategories of all orders. Use it to print the names and segments of those customers who ordered more than 20

-- pens and art supplies products.

--

-- Union, Union all, Intersect and Minus

-- 1. Combine the order numbers for orders and order ids for all shipments in a single column.

-- 2. Return non-duplicate order numbers from the orders and shipping tables in a single column.

-- 3. Find the shipment details of products with no information on the product base margin.

-- 4. What are the two most and the two least profitable products?

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-- Module: Advanced SQL

-- Session: Window Functions

-- Window Functions in Detail

-- 1. Rank the orders made by Aaron
Smayling in the decreasing order of the
resulting sales.

-- 2. For the above customer, rank the
orders in the increasing order of the
discounts provided. Also display the
-- dense ranks.

-- 3. Rank the customers in the
decreasing order of the number of orders
placed.

-- 4. Create a ranking of the number of
orders for each mode of shipment based on
the months in which they were
-- shipped.

--

-- Named Windows

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-- 1. Rank the orders in the increasing
order of the shipping costs for all
orders placed by Aaron Smayling. Also
-- display the row number for each order.
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-- Frames
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-- 1. Calculate the month-wise moving
average shipping costs of all orders
shipped in the year 2011.
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-- Session: Programming Constructs in
Stored Functions and Procedures
-- IF Statements
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-- 1. Classify an order as 'Profitable'
or 'Not Profitable'.
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-- CASE Statements

-- 1. Classify each market fact in the following ways:

-- Profits less than -500: Huge loss

-- Profits between -500 and 0:

Bearable loss

-- Profits between 0 and 500: Decent profit

-- Profits greater than 500: Great profit

-- 2. Classify the customers on the following criteria (TODO)

-- Top 20% of customers: Gold

-- Next 35% of customers: Silver

-- Next 45% of customers: Bronze

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-- Stored Functions

-- 1. Create and use a stored function to classify each market fact in the following ways:

-- Profits less than -500: Huge loss

-- Profits between -500 and 0:

Bearable loss

-- Profits between 0 and 500: Decent

profit

-- Profits greater than 500: Great
profit

--

-- Stored Procedures

-- 1. Classify each market fact in the
following ways:

-- Profits less than -500: Huge loss

-- Profits between -500 and 0:

Bearable loss

-- Profits between 0 and 500: Decent
profit

-- Profits greater than 500: Great
profit

-- The market facts with ids '1234',
'5678' and '90' belong to which
categories of profits?

--

-- Outer Join

-- 1. Return the order ids which are

present in the market facts table

-- Execute the below queries before solving the next question.

```
create table manu(  
    Manu_Id int primary key,  
    Manu_name varchar(30),  
    Manu_city varchar(30),  
);
```

```
insert into manu values  
(1, 'Navneet', 'Ahemdabad'),  
(2, 'Wipro', 'Hyderabad'),  
(3, 'Furlanco', 'Mumbai');
```

```
alter table Prod_Dimen  
add column Manu_Id int;
```

```
update Prod_dimen  
set Manu_Id = 2  
where Product_Category = 'technology';
```

-- 2. Display the products sold by all the manufacturers using both inner and outer joins

-- 3. Display the number of products sold by each manufacturer